

Somya Agarwal

somyaagarwal0608@gmail.com | 8449919761 | LinkedIn | GitHub | LeetCode

EDUCATION

Vellore Institute of Technology

Bachelor of Technology - Computer Science and Engineering

October 2022 – May 2026

CGPA: 8.63

WORK EXPERIENCE

Machine Learning Intern, Suvidha Foundation

May 2025 – July 2025

- Engineered the design and implementation of custom pipeline integrating source analysis, **credibility filtering**, and **BERT-based summarization** models, which flagged over 500 unreliable news articles per day.
- Achieved a **+35%** improvement in trust score and reduced fake news content in summaries by over 50% through rigorous evaluation.
- Built a source analysis pipeline using **credibility analysis** and **BERT-based summarization** models, decreasing the average summarization processing time by **15%**, enabling faster content assessment.

Software Developer Intern, Deeporion Technology

October 2024 – December 2025

- Developed a real-time sentiment analysis engine (**AngularJS, Java, and MySQL**), processing 25k+ customer reviews daily; reduced feedback loop by **40%**.
- Improved **MongoDB** and **PostgreSQL** queries across 10+ apps, reducing retrieval time by 35 seconds using optimized indexing and joins.
- Collaborated with cross-functional teams to deliver backend services in an agile environment, demonstrating ownership and bias for action.

SKILLS AND EXPERTISE

Languages	: Python, JAVA, HTML5, CSS3, JavaScript
Databases	: MySQL, PostgreSQL, SQLite, Oracle DB, MongoDB
Tools and Technologies	: Generative AI, LLMs, RAG, Docker, Git/GitHub CI/CD
Frameworks and Library	: GIT, REST, Spring Boot, Node.js, React.js, RESTful APIs
CS Fundamentals	: Data Structures, Algorithms, OOP, Operating Systems

ACADEMIC PROJECTS

CrisisWatch – AI-Powered Crisis Mapping Platform

- Designed a full-stack crisis mapping platform with **React, FastAPI, and PostgreSQL/POSTGIS**, enabling visualization of real-time disaster incidents with **1.5-3 seconds latency**.
- Implemented a robust **PostgreSQL + Alembic migration system**, ensuring 100% database scheme version control and smooth **CI/CD** pipelines.

AI-Powered Enterprise Knowledge Base and RAG Chatbot

- Built an end-to-end document ingestion and **metadata enrichment pipeline**, combining user-provided column descriptions with **LLM-based semantic analysis**.
- Designed and implemented an **AI-Powered Knowledge base system** to convert structured CSV datasets into semantically searchable intelligence using **LLMs** and **Retrieval-Augmented Generation (RAG)**.

Carbon NFT Marketplace

- Structured a full-stack **decentralized marketplace** where users earn blockchain-based NFT badges for purchasing eco-friendly products, integrating sustainability with Web3.
- Built and deployed **ERC-721 smart contracts** on the Polygon Testnet using Solidity and Hardhat, enabling secure NFT minting tied to carbon footprint reduction.

CERTIFICATIONS

- Full Stack Development from Apna college | Coursera Frontend verified from John Hopkin's University
- GEN AI (IBM) | Java Developer Professional (OCP)

ACHIEVEMENTS

Smart India Hackathon

- Led a team of 4 in developing a prototype ranked in top 5 out of 200 submissions; contributed to backend and ML modules.

HackOnBlocks

- Directed a standout prototype ranked as one of the top 5 projects out of 150+ teams, demonstrating strong designs and execution.